



Method of Mathematical Physics

数学物理方法

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1 复势

解析函数实部与虚部的相互求解方法

已知解析函数的实部或者虚部, 利用 Cauchy-Riemann 方程

$$\begin{cases} \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \\ \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} \end{cases} \quad \begin{cases} \frac{\partial u}{\partial \rho} = \frac{1}{\rho} \frac{\partial v}{\partial \varphi} \\ \frac{\partial v}{\partial \rho} = -\frac{1}{\rho} \frac{\partial u}{\partial \varphi} \end{cases}$$

求出虚部或者实部.

偏积分法

1. 已知 $u = u(x, y)$, 由 $\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$ 求出

$$v(x, y) = - \int \frac{\partial u}{\partial y} dx + c(y)$$

2. 求偏导

$$\frac{\partial v}{\partial y} = -\frac{\partial}{\partial y} \int \frac{\partial u}{\partial y} dx + c'(y) = \frac{\partial u}{\partial x}$$

3. 偏积分

$$c(y; C) = \int c'(y) dy + C$$

4. 由此解出 $v = v(x, y; C)$ 与 $f(z) = u(x, y) + iv(x, y; C)$.

等势线族法

已知等势线族 $\phi(x, y) = C$. 若 $\phi(x, y)$ 为调和函数, 则令 $u = \phi(x, y)$ 可求出 $f(z)$.

若 $\nabla^2 \phi \neq 0$, 则设 $u = F(\phi)$

$$\begin{cases} \frac{\partial^2 u}{\partial x^2} = F''(\phi) \left(\frac{\partial \phi}{\partial x} \right)^2 + F'(\phi) \frac{\partial^2 \phi}{\partial x^2} \\ \frac{\partial^2 u}{\partial y^2} = F''(\phi) \left(\frac{\partial \phi}{\partial y} \right)^2 + F'(\phi) \frac{\partial^2 \phi}{\partial y^2} \end{cases}$$

$$\implies F''(\phi) \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 \right] + F'(\phi) \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) = 0$$

$$\xRightarrow{\text{解出}} u = F(\phi; C_1, C_2) \implies f(z) = u(x, y; C_1, C_2) + iv(x, y; C_1, C_3)$$

Eg. 1

已知 $u(x, y)$, 求 $f(z) = u + iv$.

1. $u(x, y) = x^2 - y^2$

$$f(z) = z^2 + iC$$

2. $u = \frac{x^2 - y^2}{(x^2 + y^2)^2}, f(\infty) = 0$

$$f(z) = \frac{1}{z^2}$$

3. $u = e^x \sin y$

$$f(z) = e^z + iC$$

4. $u = e^x(x \cos y - y \sin y), f(0) = 0$

$$f(z) = ze^z$$

5. $u = \ln \rho, f(1) = 0$

$$f(z) = \ln z$$

6. $u = \varphi, f(1) = 0$

$$f(z) = -i \ln z$$

Eg. 2

已知等势线组 $x^2 + y^2 = C$, 求复势.

$$f(z) = 2C_1 \ln z + C_2 + iC_3$$

Eg. 3

求线电荷密度 λ 的均匀直线产生的静电场的复势.

$$f(z) = -\frac{\lambda}{2\pi\epsilon_0} \ln z + iC$$

2 幂级数

Eg.1 Taylor 展开

1. 在 $z_0 = 0$ 的邻域把 $f(z) = e^z$ 展开.

$$e^z = \sum_{k=0}^{\infty} \frac{z^k}{k!} \quad (|z| < \infty)$$

2. 在 $z_0 = 0$ 的邻域把 $f(z) = \sin z$ 和 $g(z) = \cos z$ 展开.

$$\sin z = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} z^{2k+1} \quad \cos z = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} z^{2k} \quad (|z| < \infty)$$

3. 在 $z_0 = 0$ 的邻域把 $f(z) = \sin^2 z$ 展开.

$$\sin^2 z = \sum_{k=1}^{\infty} \frac{(-1)^{k+1} 2^{2k-1}}{(2k)!} z^{2k} \quad (|z| < \infty)$$

4. 在 $z_0 = 0$ 的邻域把 $f(z) = \cos^2 z$ 展开.

5. 在 $z_0 = i$ 的邻域把 $f(z) = \sin z$ 展开.

6. 将 $\frac{z^2}{(z+1)^2}$ 按 $(z-1)$ 的正幂展开.

$$\frac{z^2}{(z+1)^2} = 1 + \sum_{k=0}^{\infty} \frac{(-1)^k (k-3)}{2^{k+2}} (z-1)^k \quad (|z-1| < 2)$$

7. 将 $\frac{z}{z+2}$ 按 $(z-1)$ 的正幂展开.

$$\frac{z}{z+2} = 1 - \frac{2}{3} \sum_{k=0}^{\infty} \left(-\frac{1}{3}\right)^k (z-1)^k \quad (|z-1| < 3)$$

8. 将 $\frac{z-1}{z^3}$ 按 $(z+1)$ 的正幂展开.

$$\frac{z-1}{z^3} = \sum_{k=0}^{\infty} \frac{k^2 + 5k + 4}{2} (z+1)^k \quad (|z+1| < 1)$$

Eg.2 Laurent 展开

1. 将 $\frac{1}{(z-1)(z-2)}$ 在以下区域展开:

(a) $z = 0$ 的邻域

$$\frac{1}{(z-1)(z-2)} = \sum_{k=0}^{\infty} \left(1 - \frac{1}{2^{k+1}}\right) z^k$$

(b) 以 $z = 0$ 为中心

$$\frac{1}{(z-1)(z-2)} = \begin{cases} \sum_{k=0}^{\infty} \left(1 - \frac{1}{2^{k+1}}\right) z^k & (|z| < 1) \\ -\sum_{k=0}^{\infty} \left(\frac{1}{z^{k+1}} + \frac{z^k}{2^{k+1}}\right) & (1 < |z| < 2) \\ \sum_{k=0}^{\infty} \frac{2^k - 1}{z^{k+1}} & (|z| > 2) \end{cases}$$

(c) 奇点的去心邻域

$$\frac{1}{(z-1)(z-2)}$$

(d) 以奇点为中心

$$\frac{1}{(z-1)(z-2)}$$

2. 将 $\frac{1}{(z-a)(z-b)}$ 在 $z = a$ 的去心邻域展开.

3. 将 $\frac{z}{(z-1)(z-2)^2}$ 以 $z = 0$ 为中心展开.

$$\frac{z}{(z-1)(z-2)^2} = \begin{cases} \sum_{k=0}^{\infty} \left(\frac{k+2}{2^{k+1}} - 1\right) z^k & (|z| < 1) \\ \sum_{k=0}^{\infty} \frac{k+2}{2^{k+1}} z^k + \frac{1}{z^{k+1}} & (1 < |z| < 2) \\ \sum_{k=0}^{\infty} \frac{1-2^k}{z^{k+1}} + \frac{(k+1)2^{k+1}}{z^{k+2}} & (|z| > 2) \end{cases}$$

4. 将 $\frac{z^2 - 2z + 5}{(z^2 + 1)(z - 2)}$ 在 $1 < |z| < 2$ 展开.

3 留数定理

留数定理求围道积分

设有界区域 D 的边界 C 为分段光滑的简单闭合曲线, 若除有限个孤立奇点 b_k 外, $f(z)$ 在 D 内解析, 在 $\bar{D}$ 连续, 则

$$\oint_C f(z) dz = 2\pi i \sum_{k=1}^n \operatorname{res} f(b_k) \quad (k = 1, \dots, n)$$

其中, $f(z)$ 在 b_k 处的留数为

$$\operatorname{res} f(b_k) = a_{-1}^{(k)}$$

一阶极点

$$\operatorname{res} f(b) = \lim_{z \rightarrow b} (z - b)f(z)$$

m 阶极点处的留数

$$\operatorname{res} f(b) = \frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}} [(z-b)^m f(z)]_{z=b}$$

全平面的留数定理

若 $f(z)$ 在复平面 $\mathbb{C}$ 内除了有限多个孤立奇点 b_k ($k = 1, \dots, n$) 外处处解析, 则 $f(z)$ 在 $\mathbb{C}$ 上的所有孤立奇点的留数和为 0, 即

$$\operatorname{res} f(\infty) + \sum_{k=1}^n \operatorname{res} f(b_k) \equiv 0$$

Eg. 1

1. $\oint_l \frac{dz}{(z^2 + 1)(z - 1)^2} = -\frac{1}{2}\pi i, l: x^2 + y^2 - 2x - 2y = 0.$
2. $\oint_{|z|=n} \tan \pi z dz = -4\pi i.$
3. $\oint_l \frac{dz}{(z - 3)(z^5 - 1)}, l: |z| = 2.$
4. $\oint_l \frac{dz}{z^4 + 1}, l: x^2 + y^2 = 2x.$

小圆弧引理

若 $f(z)$ 在 $\dot{U}(a, \delta)$ 内连续, 且在 $\theta_1 \leq \arg(z-a) \leq \theta_2$ 中, 当 $|z-a| \rightarrow 0$ 时, $(z-a)f(z) \Rightarrow k$, 则

$$\lim_{\delta \rightarrow 0} \int_{C_\delta} f(z) dz = ik(\theta_2 - \theta_1)$$

大圆弧引理

若 $f(z)$ 在 ∞ 的邻域连续, 在 $\theta_1 \leq \arg(z) \leq \theta_2$ 中, 当 $|z| \rightarrow \infty$ 时, $zf(z) \Rightarrow K$, 则

$$\lim_{R \rightarrow \infty} \int_{C_R} f(z) dz = iK(\theta_2 - \theta_1)$$

Jordan 引理

若在 $0 \leq \arg z \leq \pi$ 范围内, 当 $|z| \rightarrow \infty$, $Q(z) \Rightarrow 0$, 则

$$\lim_{R \rightarrow \infty} \int_{C_R} Q(z) e^{ipz} dz = 0 \quad (p > 0)$$

其中 C_R 是以原点为圆心, 半径为 R 的上半圆弧.

有理三角函数的积分

$$R(\sin \theta, \cos \theta) = R\left(\frac{z^2 - 1}{2iz}, \frac{z^2 + 1}{2z}\right) = f(z)$$

其中 R 是由 $\sin \theta, \cos \theta$ 的多项式相除得到的函数. 若 R 在 $[0, 2\pi]$ 上是连续的, 则

$$I = \int_0^{2\pi} R(\sin \theta, \cos \theta) d\theta = \oint_{|z|=1} f(z) \frac{dz}{iz} = 2\pi \sum_{|z|=1} \operatorname{res} \left\{ \frac{f(z)}{z} \right\}$$

若 R 在 $[0, 2\pi]$ 上有瑕点, 则 $f(z)$ 在 $|z|=1$ 上有奇点. 若这些奇点 β_k 均为一阶极点, 则

$$I = 2\pi \sum_{|z|=1} \operatorname{res} \left\{ \frac{f(z)}{z} \right\} + \pi \sum_k \operatorname{res} \left\{ \frac{f(z)}{z} \right\}_{z=\beta_k}$$

Eg. 2

$$1. \int_0^{2\pi} \frac{dx}{1 + \varepsilon \cos x} = \frac{2\pi}{\sqrt{1 - \varepsilon^2}}, \quad 0 < \varepsilon < 1.$$

$$2. \int_0^\pi \frac{a dx}{a^2 + \sin^2 x} = \frac{\pi}{\sqrt{a^2 + 1}}, \quad a > 0.$$

$$3. \int_0^{2\pi} \frac{d\theta}{1 + \cos^2 \theta} = \sqrt{2}\pi.$$

$$4. \int_0^{\frac{\pi}{2}} \frac{d\theta}{a + \sin^2 \theta} = \frac{\pi}{2\sqrt{a^2 + a}}, a > 0.$$

无穷积分

若 $f(z)$ 在上半平面除了有限多个孤立奇点 b_k 外解析, 在实轴上无奇点, 且在 $0 \leq \arg z \leq \pi$ 范围内, 当 $|z| \rightarrow \infty, zf(z) \rightarrow 0$, 则

$$\int_{-\infty}^{\infty} f(x) dx = 2\pi i \sum_k \operatorname{res} f(b_k)$$

Eg. 3

$$1. \int_{-\infty}^{\infty} \frac{dx}{(1+x^2)^n} = \frac{\pi(2n-2)!}{2^{2n-2}[(n-1)!]^2}, n \in \mathbb{N}^*.$$

$$2. \int_0^{\infty} \frac{dx}{x^4 + a^4} = \frac{\sqrt{2}\pi}{4a^3}, a > 0.$$

$$3. \int_{-\infty}^{\infty} \frac{x^{2m}}{1+x^{2n}} dx = \frac{\pi}{n \sin \frac{(2m+1)\pi}{2n}}, m < n \in \mathbb{N}^*.$$

$$4. \int_{-\infty}^{\infty} \frac{1+x^2}{1+x^4} dx = \sqrt{2}\pi.$$

含三角函数的无穷积分

若 $f(z)$ 在上半平面除了有限多个孤立奇点 b_k 外解析, 在实轴上无奇点, 且在 $0 \leq \arg z \leq \pi$ 范围内, 当 $|z| \rightarrow \infty, f(z) \rightarrow 0$, 则

$$\int_{-\infty}^{\infty} f(x) \cos px dx = \operatorname{Re} \left\{ 2\pi i \sum_k \operatorname{res} \left\{ f(z) e^{ipz} \right\}_{z=b_k} \right\}$$

$$\int_{-\infty}^{\infty} f(x) \sin px dx = \operatorname{Im} \left\{ 2\pi i \sum_k \operatorname{res} \left\{ f(z) e^{ipz} \right\}_{z=b_k} \right\}$$

Eg. 4

$$1. \int_0^{\infty} \frac{\cos mx}{x^2 + a^2} dx = \frac{\pi}{2a} e^{-ma}, a > 0, m > 0.$$

$$2. \int_0^{\infty} \frac{x \sin mx}{(a^2 + x^2)^2} dx = \frac{\pi m}{4a} e^{-ma}, a > 0, m > 0.$$

积分路径上存在奇点情形

若积分围道上有奇点 c , 所围区域内有有限多个孤立奇点 b_k , 则

$$\int_{-\infty}^{\infty} f(x) dx = 2\pi i \sum_k \operatorname{res} f(b_k) - \lim_{\delta \rightarrow 0} \int_{C_\delta} f(z) dz - \lim_{R \rightarrow \infty} \int_{C_R} f(z) dz$$

圆弧 C_δ 上积分值的极限可用小圆弧引理计算, C_R 上积分值的极限可用大圆弧引理或 Jordan 引理计算.

Eg. 5

1. $\int_0^{\infty} \frac{\sin x}{x} dx = \frac{\pi}{2}.$
2. $\int_0^{\infty} \frac{\sin x}{x(x^2 + 1)^2} dx = \frac{\pi}{2} - \frac{3\pi}{4e}.$
3. $\int_{-\infty}^{\infty} \frac{dx}{x^4 - 1}.$
4. $\int_{-\infty}^{\infty} \frac{\sin^2 x}{x^2} dx.$
5. $\int_{-\infty}^{\infty} \frac{\sin^3 x}{x^3} dx.$
6. $\int_{-\infty+ib}^{\infty+ib} e^{-az^2} dz, a, b > 0.$

4 δ 函数

δ 函数的特性

$$\delta(x - x_0) = \begin{cases} 0 & x \neq x_0 \\ \infty & x = x_0 \end{cases} \quad \int_a^b \delta(x - x_0) dx = \begin{cases} 1 & a < x_0 < b \\ 0 & \text{otherwise} \end{cases}$$

三维空间中的 δ 函数

$$\delta(\mathbf{r} - \mathbf{r}_0) = \delta(x - x_0)\delta(y - y_0)\delta(z - z_0)$$

Eg. 1 点电荷的电势方程

(x_0, y_0, z_0) 处有点电荷 q , 电势 $V(x, y, z)$ 满足的方程为

$$\nabla^2 V(x, y, z) = -\frac{\rho}{\varepsilon_0} = -\frac{q\delta(x - x_0)\delta(y - y_0)\delta(z - z_0)}{\varepsilon_0}$$

δ 函数的基本性质

1. $\delta(x)$ 是偶函数, $\delta'(x)$ 是奇函数.

$$2. H(x) = \int_{-\infty}^x \delta(t) dt = \begin{cases} 0 & x < 0 \\ 1 & x > 0 \end{cases} \implies \delta(x) = \frac{dH(x)}{dx}.$$

3. 对于定义在 $(-\infty, \infty)$ 的连续函数 $f(\tau)$, 有

$$\int_{-\infty}^{\infty} f(\tau)\delta(\tau - \tau_0) d\tau = f(\tau_0)$$

4. 三维空间中的挑选性

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{y}{x^2 + z} \cos(x + z)\delta(x, y - 1, z + 2) dx dy dz = -\frac{1}{2} \cos 2$$

Eg. 2 物理量的表示

1. 质量密度 $\rho_m = m\delta(\mathbf{r} - \mathbf{r}_0)$

2. 电荷密度 $\rho_e = Q\delta(\mathbf{r} - \mathbf{r}_0)$

3. 瞬时力 $F = I\delta(t - t_0)$

4. 脉冲电流 $i(t) = q\delta(t - t_0)$

δ 函数的导数性质

对任意连续函数 $f(x)$, 有

$$\int_{-\infty}^{\infty} f(x) \delta^{(n)}(x-a) dx = (-1)^n f^{(n)}(a)$$

Eg. 3

$$\int_{-1}^1 x e^x \delta'(x) dx = -1$$

δ 的复合函数性质

若 δ 函数以 $\phi(x)$ 为宗量, 则有

$$\delta(\phi(x)) = \sum_{i=1}^k \frac{\delta(x-x_i)}{|\phi'(x_i)|}$$

其中 x_i 为 $\phi(x) = 0$ 的单根. **Pf** $\delta(x^2 - a^2) = \frac{1}{2|a|} [\delta(x-a) + \delta(x+a)] = \frac{1}{2|x|} [\delta(x-a) + \delta(x+a)]$.

δ 函数的缩放性质

$$\delta(ax) = \frac{1}{|a|} \delta(x)$$

$\delta(x)$ 的 Fourier 变换

$$\delta(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\omega x} d\omega$$

Eg. 4

1. 证明

$$\delta(x) = \lim_{a \rightarrow \infty} \frac{\sin ax}{\pi x}$$

2. 利用 Fourier 变换证明

$$\delta(x) = \delta(-x) \quad \delta'(-x) = -\delta'(x)$$

3. 由 $\delta(x)$ 的积分式, 求

$$\mathcal{F} \{e^{iax}\} = 2\pi \delta(a - \omega)$$

$$\mathcal{F} \{\cos ax\} = \pi [\delta(a - \omega) + \delta(a + \omega)]$$

$$\mathcal{F}\{\sin ax\} = \pi i [\delta(a + \omega) - \delta(a - \omega)]$$

QM 中的重要公式

$$\int_{-\infty}^{\infty} e^{ikx} dk = 2\pi\delta(x)$$
$$\int_{\infty} e^{i\mathbf{k}\cdot\mathbf{r}} d^3k = (2\pi)^3\delta(\mathbf{r})$$

Eg. 5

已知 $\psi_{\mathbf{p}}(\mathbf{r}) = \frac{1}{(2\pi\hbar)^{\frac{3}{2}}} e^{\frac{i}{\hbar}\mathbf{p}\cdot\mathbf{r}}$, 证明

$$\int_{\infty} \psi_{\mathbf{p}'}^*(\mathbf{r})\psi_{\mathbf{p}}(\mathbf{r}) d^3r = \delta(\mathbf{p} - \mathbf{p}')$$

5 齐次方程、齐次边界条件的分离变量

齐次波动方程

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = 0 \\ u|_{x=0} = u|_{x=l} = 0 \\ u|_{t=0} = \phi(x), \quad \left. \frac{\partial u}{\partial t} \right|_{t=0} = \psi(x) \end{cases}$$

$$u(x, t) = X(x)T(t) \Rightarrow XT'' - a^2 X''T = 0 \Rightarrow \frac{X''}{X} = \frac{T''}{a^2 T} = -\lambda \Rightarrow \begin{cases} X'' + \lambda X = 0 \\ T'' + \lambda a^2 T = 0 \end{cases}$$

$$\begin{cases} \lambda = 0 \Rightarrow X = Ax + B \Rightarrow A = B = 0 \\ \lambda < 0 \Rightarrow X = Ae^{\sqrt{-\lambda}x} + Be^{-\sqrt{-\lambda}x} \Rightarrow A = B = 0 \\ \lambda > 0 \Rightarrow X = B_n \sin \frac{n\pi x}{l} \quad \lambda_n = \left(\frac{n\pi}{l}\right)^2 \Rightarrow T(t) = c_n \cos \frac{n\pi at}{l} + d_n \sin \frac{n\pi at}{l} \end{cases}$$

$$\Rightarrow u(x, t) = \sum_{n=1}^{\infty} \left(C_n \cos \frac{n\pi at}{l} + D_n \sin \frac{n\pi at}{l} \right) \sin \frac{n\pi x}{l}$$

$$u|_{t=0} = \sum_{n=1}^{\infty} C_n \sin \frac{n\pi x}{l} = \phi(x) \Rightarrow C_n = \frac{2}{l} \int_0^l \phi(x) \sin \frac{n\pi x}{l} dx$$

$$\left. \frac{\partial u}{\partial t} \right|_{t=0} = \sum_{n=1}^{\infty} \frac{n\pi a}{l} D_n \sin \frac{n\pi x}{l} = \psi(x) \Rightarrow D_n = \frac{2}{n\pi a} \int_0^l \psi(x) \sin \frac{n\pi x}{l} dx$$

Rmk 齐次输运方程的分离变量与齐次波动方程的类似. 基本思路为:

设形式解 $\rightarrow$ 利用边界条件确定本征值与本征函数 $\rightarrow$ 叠加 $\rightarrow$ 利用初始条件确定常数

极坐标系 Laplace 方程

$$\begin{cases} \frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \varphi^2} = 0 \\ u(\rho, \varphi) = u(\rho, \varphi + 2\pi) \end{cases}$$

$$u = R(\rho)\Phi(\varphi) \Rightarrow \begin{cases} \Phi'' + \lambda\Phi = 0 \Rightarrow \Phi = a_m \cos m\varphi + b_m \sin m\varphi \quad (\lambda = m^2; m \in \mathbb{N}) \\ \rho^2 R'' + \rho R' - m^2 R = 0 \xrightarrow{\rho=e^t} R = \begin{cases} c_m \rho^m + d_m \rho^{-m} & m \neq 0 \\ c_0 + d_0 \ln \rho & m = 0 \end{cases} \end{cases}$$

$$\Rightarrow u = C_0 + D_0 \ln \rho + \sum_{m=1}^{\infty} (A_m \cos m\varphi + B_m \sin m\varphi) \rho^m + (C_m \cos m\varphi + D_m \sin m\varphi) \rho^{-m}$$

$$u|_{r=0} = \text{有限值} \Rightarrow u = C_0 + \sum_{m=1}^{\infty} (A_m \cos m\varphi + B_m \sin m\varphi) \rho^m$$

Eg. 1 导体圆柱

在匀强电场中垂直于电场放入半径为 a 的无限长接地导体圆柱, 求柱内外电势分布.

$$u_1 = 0 \quad u_2 = D_0 \ln \frac{\rho}{a} - E_0 \rho \cos \varphi + E_0 \frac{a^2}{\rho} \cos \varphi$$

Eg. 2 电介质圆柱

在匀强电场中垂直于电场放入半径为 a 的无限长电介质圆柱 (相对介电常量为 ε_r), 求柱内外电势分布. (设 $u|_{\rho \rightarrow 0} = 0$)

$$u_1 = -\frac{2E_0}{\varepsilon_r + 1} \rho \cos \varphi \quad u_2 = -E_0 \rho \cos \varphi + \frac{\varepsilon_r - 1}{\varepsilon_r + 1} E_0 \frac{a^2}{\rho} \cos \varphi$$

球坐标系 Laplace 方程

$$\begin{aligned} & \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 u}{\partial \varphi^2} = 0 \\ \xrightarrow{u(r, \theta, \varphi) = R(r) \Theta(\theta) \Phi(\varphi)} & \frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \frac{1}{\Phi \sin^2 \theta} \frac{d^2 \Phi}{d\varphi^2} = 0 \end{aligned}$$

$$\xrightarrow{\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) = l(l+1)} \frac{\sin \theta}{\Theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \frac{1}{\Phi} \frac{d^2 \Phi}{d\varphi^2} + l(l+1) \sin^2 \theta = 0$$

$$\xrightarrow{\frac{1}{\Phi} \frac{d^2 \Phi}{d\varphi^2} = -m^2} \frac{\sin \theta}{\Theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + l(l+1) \sin^2 \theta = m^2$$

$$\xrightarrow{\substack{x = \cos \theta \\ y(x) = \Theta(\cos \theta)}} (1-x^2) \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + \left[l(l+1) - \frac{m^2}{1-x^2} \right] y = 0$$

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) = l(l+1) \xrightarrow{r=e^t} R = a_l r^l + b_l r^{-(l+1)}$$

$$\frac{1}{\Phi} \frac{d^2 \Phi}{d\varphi^2} = -m^2 \implies \Phi = c_m \cos m\varphi + d_m \sin m\varphi \quad (m = 0, 1, 2, \dots)$$

$m = 0$, 轴对称:

$$u = \sum_{l=0}^{\infty} (A_l r^l + B_l r^{-(l+1)}) P_l(\cos \theta)$$

$m \neq 0$, 非轴对称:

$$u = \sum_{l=0}^{\infty} \sum_{m=0}^l r^l (A_l^m \cos m\varphi + B_l^m \sin m\varphi) P_l^m(\cos \theta) + r^{-(l+1)} (C_l^m \cos m\varphi + D_l^m \sin m\varphi) P_l^m(\cos \theta)$$

6 非齐次方程、非齐次边界条件的分离变量

一维齐次边界条件对应的本征值与本征函数

边界条件	本征值	空间本征函数	n
$u _{x=0} = u _{x=l} = 0$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2$	$X_n(x) = B_n \sin \frac{n\pi x}{l}$	$n \in \mathbb{N}^*$
$\frac{\partial u}{\partial x}\bigg _{x=0} = \frac{\partial u}{\partial x}\bigg _{x=l} = 0$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2$	$X_n(x) = A_n \cos \frac{n\pi x}{l}$	$n \in \mathbb{N}$
$u _{x=0} = \frac{\partial u}{\partial x}\bigg _{x=l} = 0$	$\lambda_n = \left[\frac{(n+1/2)\pi}{l}\right]^2$	$X_n(x) = B_n \sin \frac{(n+1/2)\pi x}{l}$	$n \in \mathbb{N}$
$\frac{\partial u}{\partial x}\bigg _{x=0} = u _{x=l} = 0$	$\lambda_n = \left[\frac{(n+1/2)\pi}{l}\right]^2$	$X_n(x) = A_n \cos \frac{(n+1/2)\pi x}{l}$	$n \in \mathbb{N}$

非齐次方程的齐次化: Fourier 级数法

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t) \\ u|_{x=0} = u|_{x=l} = 0 \quad \rightarrow \text{齐次} \\ u|_{t=0} = \frac{\partial u}{\partial t}\bigg|_{t=0} = 0 \quad \rightarrow \text{齐次} \end{cases}$$

$$\Rightarrow \begin{cases} u(x, t) = \sum_{n=1}^{\infty} T_n(t) \sin \frac{n\pi x}{l} \\ f(x, t) = \sum_{n=1}^{\infty} f_n(t) \sin \frac{n\pi x}{l} \\ f_n(t) = \frac{2}{l} \int_0^l f(x, t) \sin \frac{n\pi x}{l} dx \\ T_n''(t) + \left(\frac{n\pi a}{l}\right)^2 T_n(t) = f_n(t) \end{cases}$$

非齐次方程的齐次化: 冲量定理法

- 波动方程

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t) \\ (\partial_x)u|_{x=0} = (\partial_x)u|_{x=l} = 0 \quad \rightarrow \text{齐次} \\ u|_{t=0} = \frac{\partial u}{\partial t}\bigg|_{t=0} = 0 \quad \rightarrow \text{齐次} \end{cases}$$

$$\Rightarrow \begin{cases} \frac{\partial^2 v}{\partial t^2} - a^2 \frac{\partial^2 v}{\partial x^2} = 0 \\ (\partial_x)v|_{x=0} = (\partial_x)v|_{x=l} = 0 \\ v|_{t=\tau} = 0, \quad \frac{\partial v}{\partial t}\bigg|_{t=\tau} = f(x, \tau) \end{cases}$$

• 输运方程

$$\begin{cases} \frac{\partial u}{\partial t} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t) \\ (\partial_x)u|_{x=0} = (\partial_x)u|_{x=l} = 0 \rightarrow \text{齐次} \\ u|_{t=0} = 0 \rightarrow \text{齐次} \end{cases}$$

$$\Rightarrow \begin{cases} \frac{\partial v}{\partial t} - a^2 \frac{\partial^2 v}{\partial x^2} = 0 \\ (\partial_x)v|_{x=0} = (\partial_x)v|_{x=l} = 0 \\ v|_{t=\tau} = f(x, \tau) \end{cases}$$

$$\xrightarrow{\text{解出}} v(x, t, \tau) \Rightarrow u(x, t) = \int_0^t v(x, t, \tau) d\tau$$

非齐次边界条件的齐次化：辅助函数法

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = 0 \\ (\partial_x)u|_{x=0} = \mu(t), \quad (\partial_x)u|_{x=l} = \nu(t) \\ u|_{t=0} = \phi(x), \quad \frac{\partial u}{\partial t}\bigg|_{t=0} = \psi(x) \end{cases} \xrightarrow{u=v+w} \begin{cases} (\partial_x)w|_{x=0} = \mu(t) \\ (\partial_x)w|_{x=l} = \nu(t) \end{cases}$$

$$w(x, t) = A(t)x + B(t) \quad \text{or} \quad w_{22}(x, t) = A(t)x^2 + B(t)x + C(t)$$

$$\Rightarrow \begin{cases} \frac{\partial^2 v}{\partial t^2} - a^2 \frac{\partial^2 v}{\partial x^2} = -\frac{\partial^2 w}{\partial t^2} + a^2 \frac{\partial^2 w}{\partial x^2} \\ (\partial_x)v|_{x=0} = (\partial_x)v|_{x=l} = 0 \\ v|_{t=0} = \phi(x) - w|_{t=0}, \quad \frac{\partial v}{\partial t}\bigg|_{t=0} = \psi(x) - \frac{\partial w}{\partial t}\bigg|_{t=0} \end{cases}$$

Eg. 1

有一根长为 l ，侧面绝热，初始温度为 0°C 的均匀细杆， $x=l$ 端保持 0°C ， $x=0$ 端温度随时间线性上升，即 $u|_{x=0} = ct$ ，求杆上的温度分布 $u(x, t)$ 。

$$u(x, t) = ct \left(1 - \frac{x}{l}\right) + \sum_{n=1}^{\infty} \frac{2cl^2}{n^3\pi^3a^2} \left(e^{-(\frac{n\pi a}{l})^2 t} - 1\right) \sin \frac{n\pi x}{l}$$

7 球函数

Legendre 多项式

$$P_l(x) = \sum_{k=0}^{[l/2]} (-1)^k \frac{(2l-2k)!}{2^l k! (l-k)! (l-2k)!} x^{l-2k}$$

$$\Rightarrow \begin{cases} P_{2n}(0) = (-1)^n \frac{(2n-1)!!}{(2n)!!} \\ P_{2n+1}(0) = 0 \end{cases}$$

微分表示

$$P_l(x) = \frac{1}{2^l l!} \frac{d^l}{dx^l} (x^2 - 1)^l$$

积分表示

$$P_l(x) = \frac{1}{\pi} \int_0^\pi (x + i\sqrt{1-x^2} \cos \varphi)^l d\varphi \Rightarrow \begin{cases} P_l(1) = 1 \\ P_l(-1) = (-1)^l \end{cases}$$

正交关系式

$$\int_{-1}^1 P_l(x) P_k(x) dx = \delta_{kl} \frac{2}{2l+1}$$

广义 Fourier 展开

$$f(x) = \sum_{l=0}^{\infty} f_l P_l(x) \quad f_l = \frac{2l+1}{2} \int_{-1}^1 f(x) P_l(x) dx$$

球域 Laplace 方程轴对称通解

$$u(r, \theta) = \sum_{l=0}^{\infty} \left(A_l r^l + B_l \frac{1}{r^{l+1}} \right) P_l(\cos \theta)$$

Eg. 1 均匀介质球

在匀强电场 $\mathbf{E} = E_0 \mathbf{e}_z$ 中放入均匀介质球, 半径 r_0 , 相对介电常数 ε_r , 求解球内外电势.

$$\begin{cases} \nabla^2 u_1 = 0 \\ u_1|_{r=0} = \text{有限值} \end{cases} \quad \begin{cases} \nabla^2 u_2 = 0 \\ u_2|_{r \rightarrow \infty} = -E_0 z = -E_0 r \cos \theta \end{cases} \quad \begin{cases} u_1|_{r=r_0} = u_2|_{r=r_0} \\ \varepsilon_r \frac{\partial u_1}{\partial r} \Big|_{r=r_0} = \frac{\partial u_2}{\partial r} \Big|_{r=r_0} \end{cases}$$

$$u_1 = -\frac{3}{\varepsilon_r + 2} E_0 r \cos \theta \quad u_2 = \frac{\varepsilon_r - 1}{\varepsilon_r + 2} r_0^3 E_0 \frac{1}{r^2} \cos \theta - E_0 r \cos \theta$$

Eg. 2 接地导体球

放入接地导体球, 求解球内外电势.

$$\begin{cases} \nabla^2 u_1 = 0 \\ u_1|_{r=0} = \text{有限值} \end{cases} \quad \begin{cases} \nabla^2 u_2 = 0 \\ u_2|_{r \rightarrow \infty} = -E_0 r \cos \theta \end{cases} \quad \begin{cases} u_1|_{r=r_0} = u_2|_{r=r_0} = 0 \\ \left. \frac{\partial u_1}{\partial r} \right|_{r=r_0} = 0 \end{cases}$$

$$u_1 = 0 \quad u_2 = E_0 r_0^3 \frac{1}{r^2} \cos \theta - E_0 r \cos \theta$$

Legendre 多项式的母函数

- 单位球

$$\frac{1}{\sqrt{1 - 2r \cos \theta + r^2}} = \begin{cases} \sum_{l=0}^{\infty} r^l P_l(\cos \theta) & (r < 1) \\ \sum_{l=0}^{\infty} \frac{1}{r^{l+1}} P_l(\cos \theta) & (r > 1) \end{cases}$$

- 非单位球

$$\frac{1}{\sqrt{R^2 - 2rR \cos \theta + r^2}} = \begin{cases} \sum_{l=0}^{\infty} \frac{r^l}{R^{l+1}} P_l(\cos \theta) & (r < R) \\ \sum_{l=0}^{\infty} \frac{R^l}{r^{l+1}} P_l(\cos \theta) & (r > R) \end{cases}$$

- 物理意义: 位于极轴 R 处的电荷 $q = 4\pi\varepsilon_0$ 产生的电势为

$$u = \frac{1}{\sqrt{R^2 + r^2 - 2rR \cos \theta}}$$

Eg. 3 接地导体球

在点电荷 $4\pi\varepsilon_0 q$ 产生的电场中放入接地导体球, 半径 a , 球心与点电荷相距 r_1 , 求解电势分布. ($\nabla^2 u \neq 0$)

$$u = \frac{q}{\sqrt{r_1^2 + a^2 - 2r_1 a \cos \theta}} + v \quad \begin{cases} \nabla^2 v = 0 \\ v|_{r=a} = -\frac{q}{\sqrt{r_1^2 + a^2 - 2r_1 a \cos \theta}} \\ v|_{r \rightarrow \infty} = 0 \end{cases}$$

$$u_1 = 0 \quad u_2 = \frac{q}{\sqrt{r_1^2 + a^2 - 2r_1 a \cos \theta}} - \sum_{l=0}^{\infty} q \frac{a^{2l+1}}{r_1^{l+1}} \frac{1}{r^{l+1}} P_l(\cos \theta)$$

Eg. 4 均匀介质球

放入均匀介质球, 求解电势分布.

$$\begin{cases} u_1 = \frac{q}{\sqrt{r_1^2 + a^2 - 2r_1a \cos \theta}} + v_1 \\ u_2 = \frac{q}{\sqrt{r_1^2 + a^2 - 2r_1a \cos \theta}} + v_2 \end{cases} \quad \begin{cases} \nabla^2 v_1 = 0 \\ \nabla^2 v_2 = 0 \end{cases} \quad \begin{cases} u_1|_{r=a} = u_2|_{r=a} \\ \varepsilon_r \frac{\partial u_1}{\partial r} \Big|_{r=a} = \frac{\partial u_2}{\partial r} \Big|_{r=a} \end{cases}$$

$$u_1 = \frac{q}{\sqrt{r_1^2 + a^2 - 2r_1a \cos \theta}} + \sum_{l=1}^{\infty} \frac{(1 - \varepsilon_r)ql}{(\varepsilon_r l + l + 1)r_1^{l+1}} r^l P_l(\cos \theta)$$

$$u_2 = \frac{q}{\sqrt{r_1^2 + a^2 - 2r_1a \cos \theta}} + \sum_{l=1}^{\infty} \frac{(1 - \varepsilon_r)qla^{2l+1}}{(\varepsilon_r l + l + 1)r_1^{l+1}} \frac{1}{r^{l+1}} P_l(\cos \theta)$$

连带 Legendre 函数

$$P_l^m(x) = (1 - x^2)^{\frac{m}{2}} P_l^{(m)}(x)$$

非轴对称解

$$u = \sum_{l=0}^{\infty} \sum_{m=0}^l \left[(A_{lm} \cos m\varphi + B_{lm} \sin m\varphi) r^l + (C_{lm} \cos m\varphi + D_{lm} \sin m\varphi) \frac{1}{r^{l+1}} \right] P_l^m(\cos \theta)$$

Eg. 5 非轴对称系统的电势

1. 半径为 r_0 的球形区域内部无电荷, 球面上电势为 $u_0 \sin^2 \theta \sin \varphi \cos \varphi$, 求解球域内部电势

$$\begin{cases} \nabla^2 u = 0 \\ u|_{r=r_0} = u_0 \sin^2 \theta \sin \varphi \cos \varphi \\ u|_{r=0} = \text{有限值} \end{cases}$$

$$u = \frac{u_0}{6r_0^2} r^2 P_2^2(\cos \theta) \sin 2\varphi$$

2. 在半径为 r_0 的球域外部求解

$$\begin{cases} \nabla^2 u = 0 \\ \frac{\partial u}{\partial r} \Big|_{r=r_0} = u_0 \left(\sin^2 \theta \sin^2 \varphi - \frac{1}{3} \right) \\ u|_{r \rightarrow \infty} = \text{有限值} \end{cases}$$

$$u = \frac{u_0 r_0^4}{18} \frac{1}{r^3} P_2^2(\cos \theta) \cos 2\varphi + \frac{u_0 r_0^4}{9} \frac{1}{r^3} P_2^0(\cos \theta)$$